

Module 1. Lesson 2.

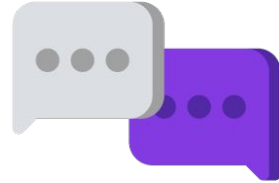
Variables

**Link to the
methodological
guidelines**



Discussion:

First professional task



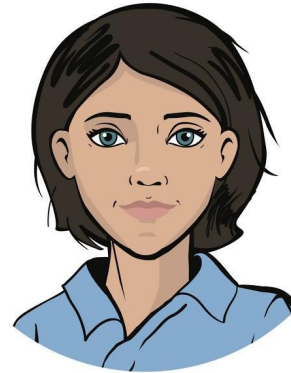
The “Travel agency calculator” project

Our development team needs help!

The ProTeam programmers are working on a big project to **automatically calculate travel costs**.

Yesterday, they got a call from the head of the company that placed the order. He asked them to help with some of the calculations *now*.

The programmers are currently busy with the project itself. Junior developers, we need your help!



Emily,
Project Manager



Discussion
of the task



The “Travel agency calculator” project

Example:

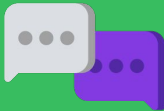
Based on the ticket price and the number of tourists, calculate the total price of a given group excursion package.

Guide's name: **Anderson**

Number of adults in the group: **18**

Price of an adult ticket:
1990

How do we perform the calculation?



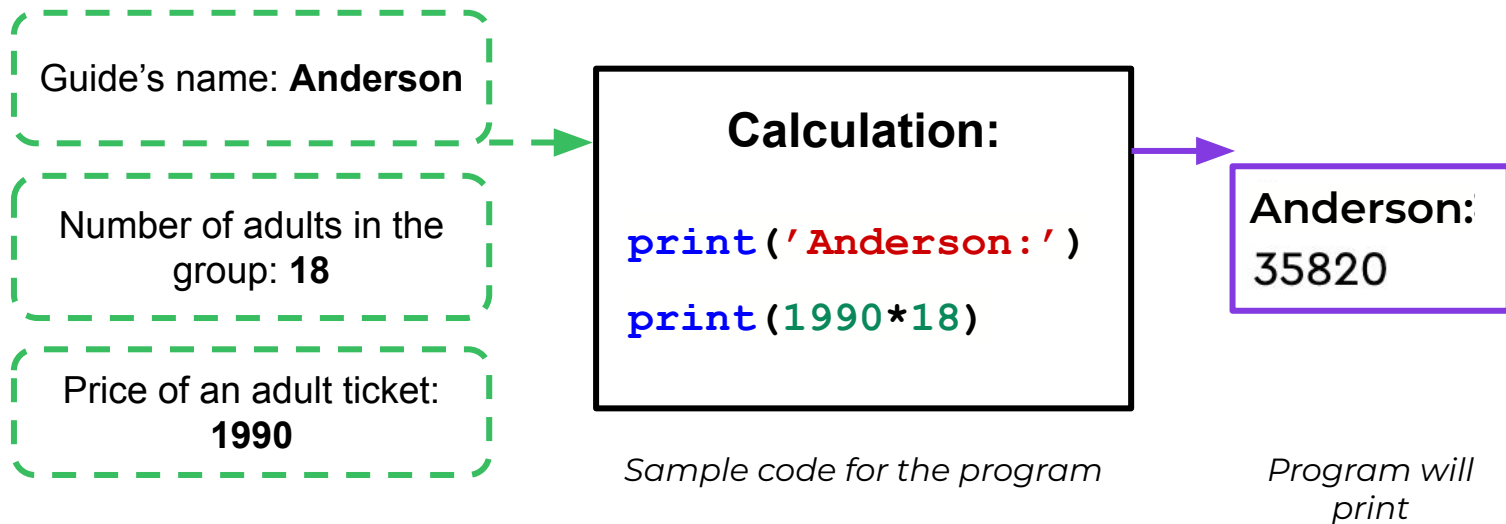
Discussion
of the task



The “Travel agency calculator” project

Example:

Based on the ticket price and the number of tourists, calculate the total price of a given group excursion package.

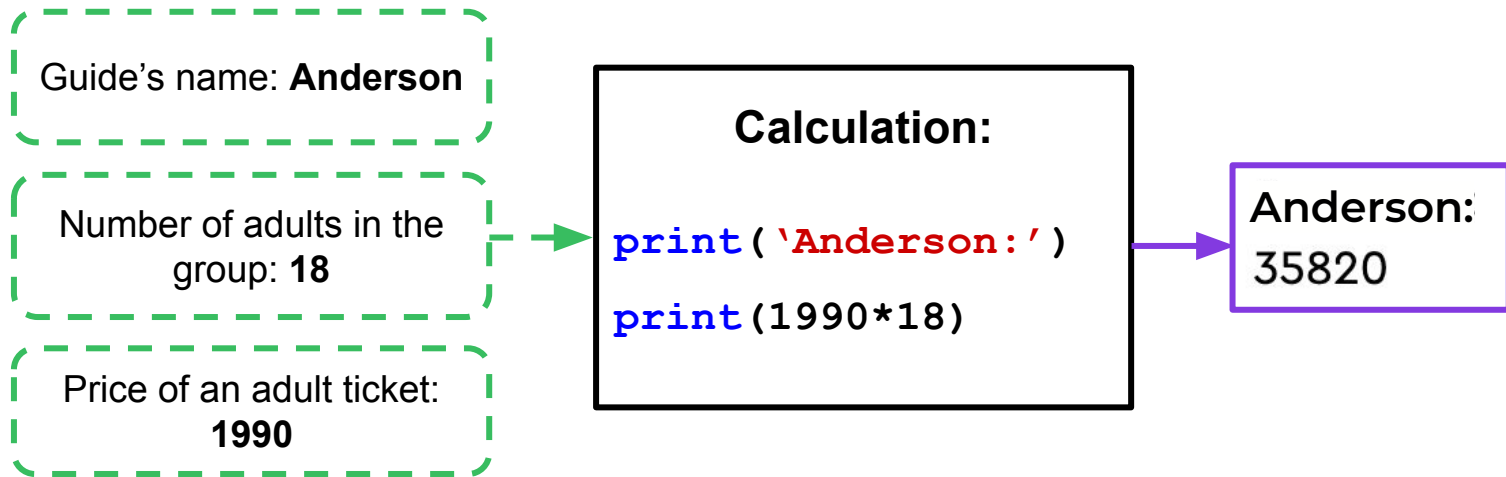


Discussion
of the task

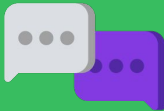


The “Travel agency calculator” project

Based on the ticket price and the number of tourists, calculate the total price of a given group excursion package.



However, the accountant is given a list of **18 different groups every day**.
Will we still be able to perform the calculations using this program?
The accountant does not know how to code.



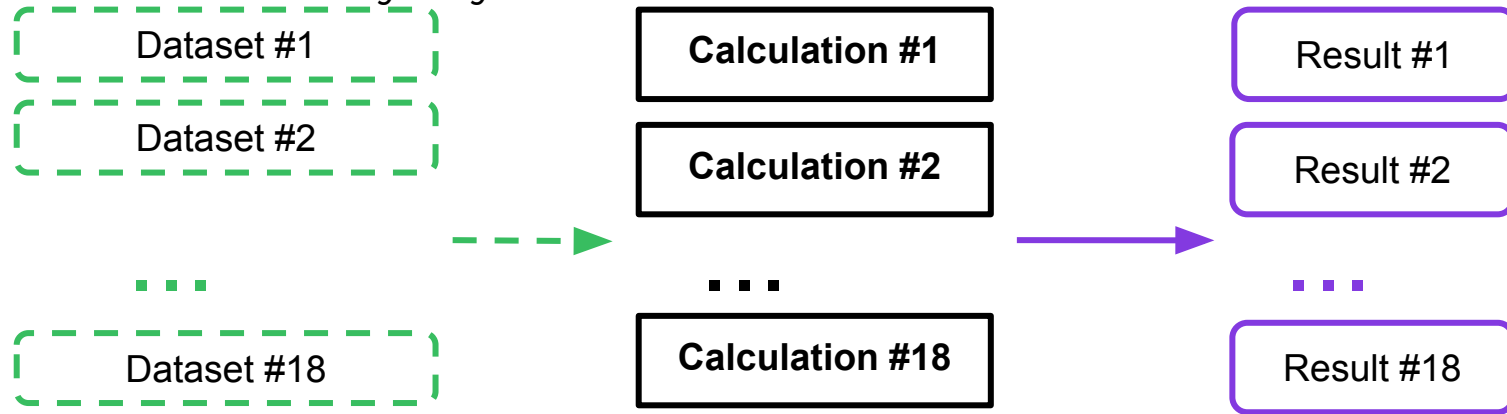
Discussion
of the task



The “Travel agency calculator” project

Now, this program only does the calculation for one specific package.

The developer in charge will have to write 18 programs for the accountant every day.



What would **improve** the program?

Can the accountant **get the work done using** just **one program**?

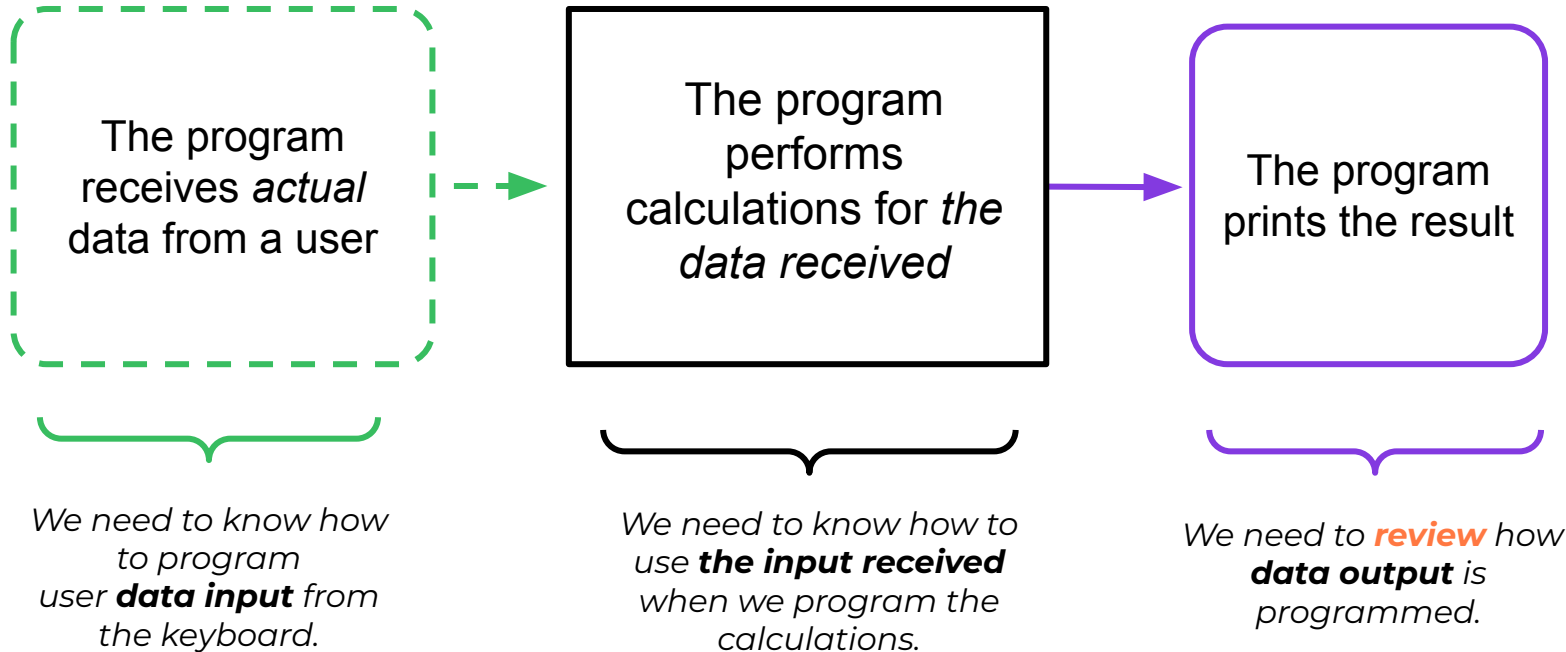


Discussion
of the task



The “Travel agency calculator” project

④ We had a universal program that was able to work with different datasets, the accountant would only need that one program.



Discussion
of the task



The goal of the workday is to

automate the calculations for the travel agency.

The client has assigned several urgent cases to ProTeam. We need to program these in a format convenient for the agency.

Today you will learn:

- that a variable is a tool that helps create universal code;
- how to program data input by users;
- one of the ways to keep track of variable values.



Discussion
of the task



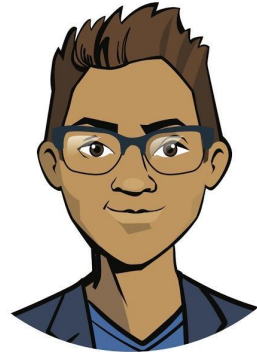
Confirmation of qualifications



A confirmation of qualifications **is a regular meeting with a lead** **programmer.**

At this meeting, you will need to demonstrate your professional knowledge and skills.

If your qualifications are confirmed, you will be allowed to work on the project.



Cole
Senior Developer



Confirmation of
qualifications



What is an algorithm, a programming language, a program ?



Confirmation of
qualifications



What is an algorithm, a programming language, a program ?

An **algorithm** is a sequence of actions to achieve a goal.

A **programming language** is a language to communicate with machines. It includes commands. Each command has a single meaning.

A **program** is an algorithm written in a programming language.



Confirmation of
qualifications



What Python language rules do you know?

?

?

?

?

?

?



Confirmation of
qualifications



What Python language rules do you know?



Rule of beginning

The first command in a program shall be written in the beginning of a line.

Rule of order

Commands shall be executed in order if they are written one below the other.

Rules of code style

In Python, replacing lowercase letters with uppercase ones is not allowed. An accidental character (even a space or comma) may break the program.

There are other rules, too. Today we will expand this list

Confirmation of qualifications



What is a **function** ? Which function do you already know?

A **function** is...



Confirmation of
qualifications



What is a **function** ? Which **function** do you already know?

A **function** is an algorithm that is composed in a programming language and has a unique name.

print() is a function for printing the parameters specified in its parentheses.

<i>Function syntax</i>	<i>Program will print</i>
<code>print('Hello, world!')</code>	Hello, world!
<code>print('Hello', 'world!')</code>	Hello, world!

Automatic space



Confirmation of
qualifications



Which parameters can **print** output?



Confirmation of
qualifications



What is specified in the parentheses when we use **print**?

As arguments, `print()` may accept words, numbers, and arithmetic expressions.

<i>Function syntax</i>	<i>Operation</i>	<i>Output</i>
<code>print(2*7)</code>	?	?
<code>print(10+4)</code>	?	?
<code>print(20-6)</code>	?	?
<code>print(28/2)</code>	?	?
<code>print(136%10)</code>	?	?



Confirmation of
qualifications



What is specified in the parentheses when we use **print**?

As arguments, `print()` may accept words, numbers, and arithmetic expressions.

<i>Function syntax</i>	<i>Value</i>	<i>Output</i>
<code>print(2*7)</code>	Multiplication	14
<code>print(10+4)</code>	Sum	14
<code>print(20-6)</code>	Difference	14
<code>print(28/2)</code>	Quotient	14.0
<code>print(136%10)</code>	Division with remainder	6



Confirmation of
qualifications



Can we set our own **order** of **operations** ?



Confirmation of
qualifications



Can we set our own **order of operations** ?

Yes, we can:

<i>Function syntax</i>	<i>Output</i>
<code>print(1+1*7)</code>	8
<code>print((1+1)*7)</code>	14

In mathematics, division, multiplication, determining the quotient and the remainder are executed first, followed by addition and subtraction.

By using parentheses, we can modify the order of operations.



Confirmation of
qualifications



Qualifications confirmed!

Great, you are ready to brainstorm and complete your work task!

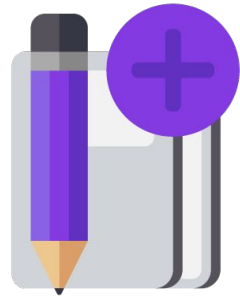


Confirmation of
qualifications



Brainstorm:

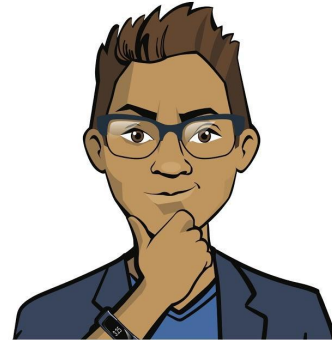
Variables



Variables

A **variable** is a data element that has its own name.

Variables are used to work with data that can change.



Brainstorm



Defining a variable

If you want to use a variable in your program, you need to:

- create a variable by giving it a **name**;
- set the variable's **value**.



Brainstorm



Defining a variable

If you want to use a variable in your program, you need to:

- create a variable by giving it a **name**;
- set the variable's **value**.

Example:

name → `hours` = `6` ← *value*
`minutes = 42`

```
print('According to statistics, in 2019, an Internet user  
      spent', hours, 'hours', minutes,  
      'minutes online every day')
```



Brainstorm

Assignment operator

To set a variable's value, you need to use the assignment operator.

Variable Value

Assignment
operator

hours = 6

print(hours)

The program will print:

6



Brainstorm



Assignment operator

The assignment operator can change the value of an existing variable.

```
hours = 6
```

```
hours = 7.5
```

↑
Changing the initial
value

```
print(hours)
```

The program will print:

7.5



Brainstorm



What name can you give to a variable?

- You **can** use letters, digits, and underscores.
- Every variable's name **must** begin with a letter or an underscore.
- A variable's name **must not** use the language's commands and other reserved names.



Brainstorm



Variable names

Compare some effective and ineffective variable names.

<i>Ineffective</i> name	Why?
<code>a = 56</code> <code>num = 1.34</code>	Other programmers will not understand what this variable stands for.
<code>number_of_students_at_school = 1108</code>	The name is too long. It is not convenient to use.



Brainstorm



Variable names

Compare some effective and ineffective variable names.

<i>Ineffective</i> name	Why?
<code>a = 56</code> <code>num = 1.34</code>	Other programmers will not understand what this variable stands for.
<code>number_of_students_at_school = 1108</code>	The name is too long. It is not convenient to use.

<i>Effective</i> name	Why?
<code>surname = 'Smith'</code> <code>ticket_price = 1999</code> <code>amount_students = 826</code>	The name explains the variable's purpose. Your fellow programmers will be able to read your program and quickly understand the point.



Brainstorm

Data types

The data hidden behind variable names can be of different types. We know three:

- **integer** numbers,
- **decimal** fractions,
- **string** data.

Different data types can be used to program different actions.



Brainstorm



Data types

The data hidden behind variable names can be of different types. We know three:

- **integer** numbers,
- **decimal** fractions,
- **string** data.

Numeric type		String type
144	<u>Integer</u> number (int)	'Calvin' (str)
48.3	Decimal fraction (float)	'256' (str)
(2*11)	<u>Integer</u> number (int)	'15.05.2007' (str)
(4*8.2)	Decimal fraction (float)	'Data received' (str)



Brainstorm

Working with numeric variables

Example. The following program calculates an employee's salary for a certain period.

What will the program print?

```
daily_salary = 1000
```

```
days = 22
```

```
total = daily_salary*days
```

```
print(total)
```



Brainstorm

Working with numeric variables

Example. The following program calculates an employee's salary for a certain period.

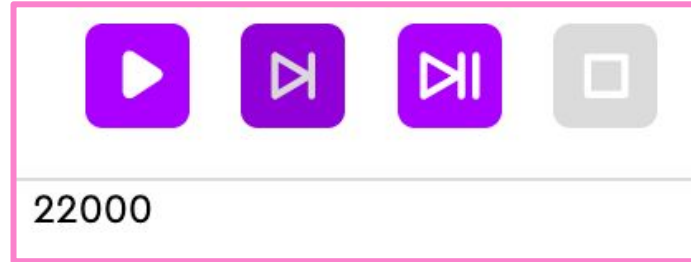
What will the program print?

```
daily_salary = 1000
```

```
days = 22
```

```
total = daily_salary*days
```

```
print(total)
```



Brainstorm



Working with numeric variables

Example. The following program calculates an employee's salary for a certain period.

An employer using this program decided to trick one of their employees and **change the daily pay**. What will the program print?

```
daily_salary = 1000
```

```
days = 22
```

```
daily_salary = 500
```

```
total = daily_salary*days
```

```
print(total)
```



Brainstorm

Working with numeric variables

Example. The following program calculates an employee's salary for a certain period.

An employer using this program decided to trick one of their employees and **change the daily pay**. What will the program print?

```
daily_salary = 1000
```

```
days = 22
```

```
daily_salary = 500
```

```
total = daily_salary*days
```

```
print(total)
```



11000

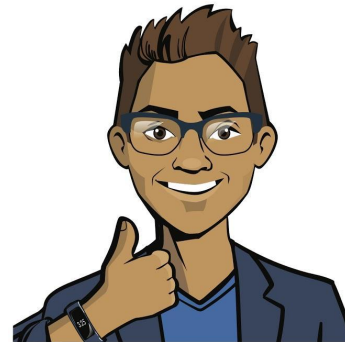


Brainstorm



Conclusions:

1. A variable is a data element that has its own name.
2. The assignment operator is used to set a variable's initial value or change its current value.
3. The data behind variables can be of different types. For now, we know that there are numeric and string types.



Brainstorm

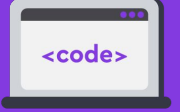
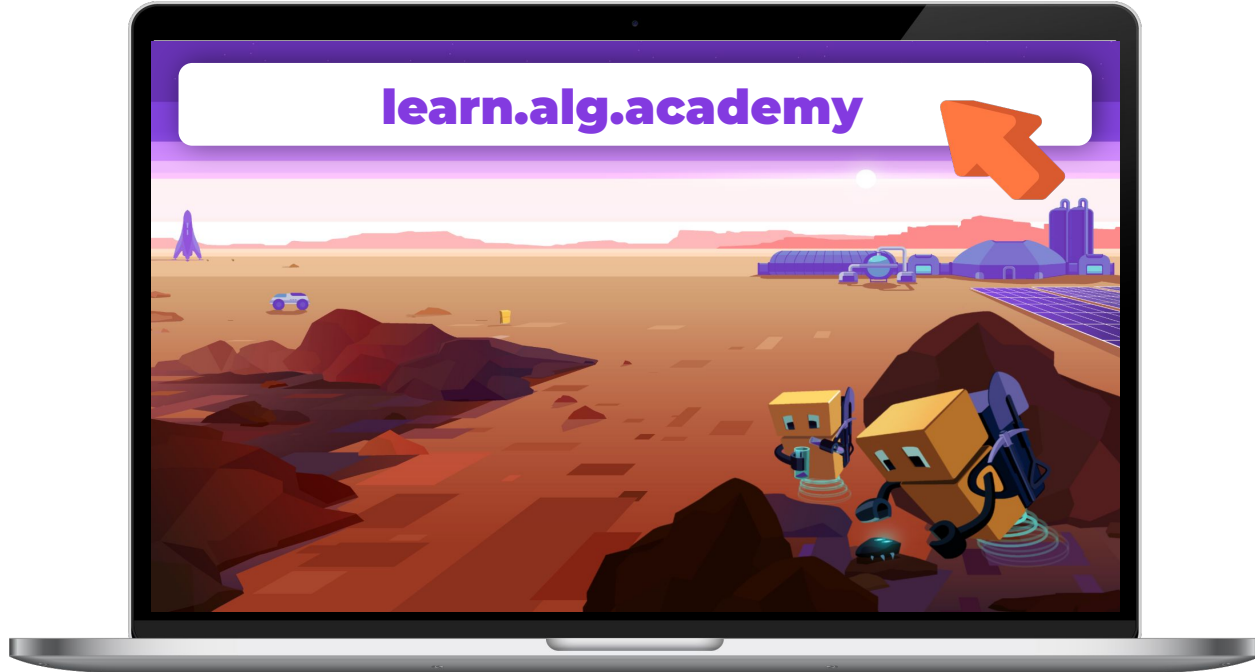


Travel agency: Automating the calculations



Do the tasks on the platform

➡ “Travel agency: calculations”



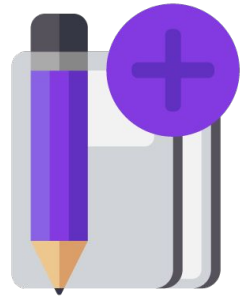
Travel agency:
calculations

Break



Brainstorm:

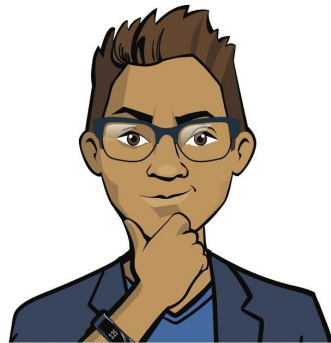
Data input



A broken program

We have just received an email from the accountant of the travel agency you have been working with.

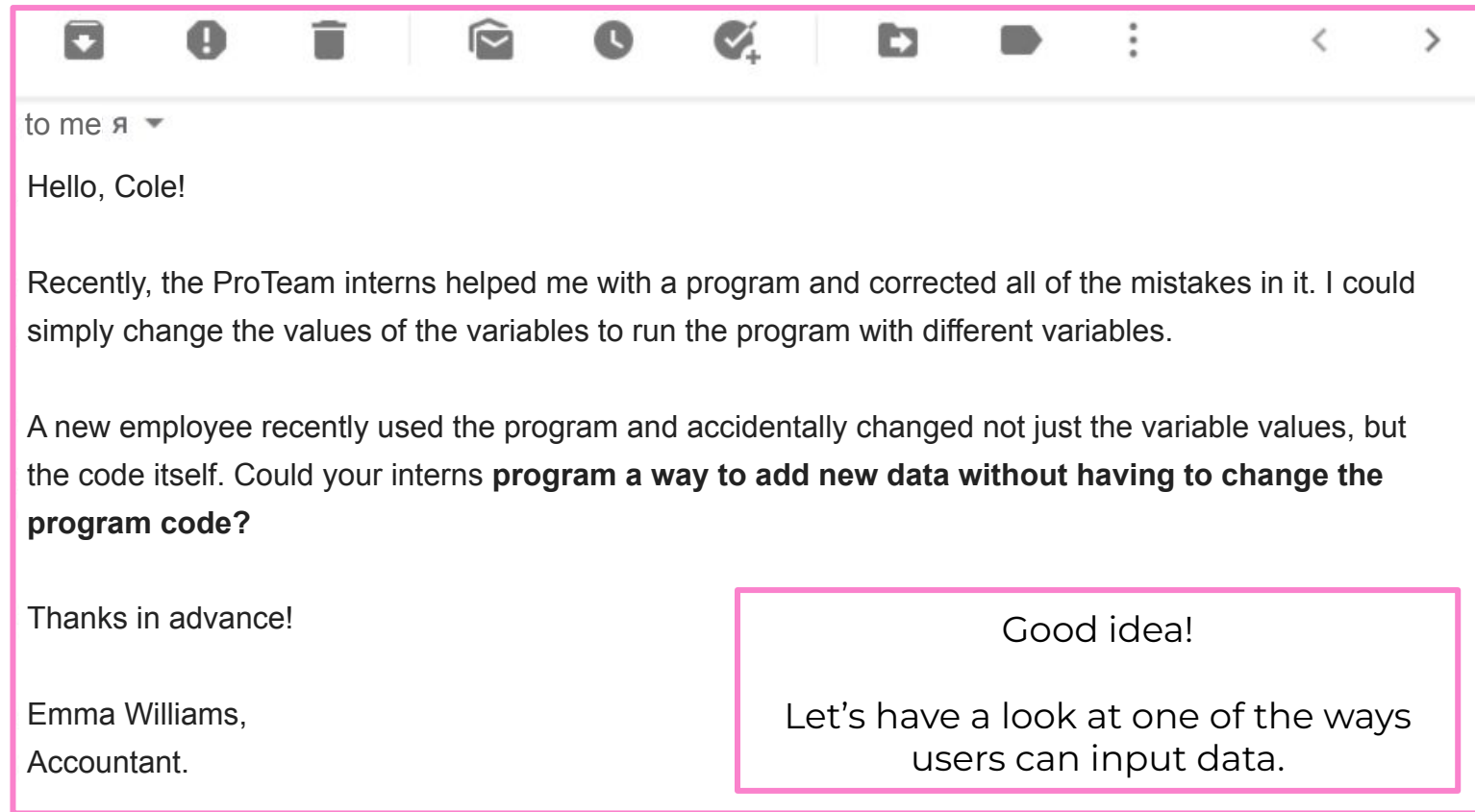
They seem to be concerned about something.



Brainstorm



A broken program



Brainstorm

The input() function

`input()` is a function to input data from the keyboard.

```
surname = input('Enter the manager's surname:')  
city = input('Enter the city:')  
print('The branch manager in', city, ' is', surname)
```

What will the program print?



Brainstorm

The input() function

`input()` is a function to input data from the keyboard.

```
surname = input('Enter the manager's surname:')  
city = input('Enter the city:')  
print('The branch manager in', city, ' is', surname)
```



Enter the manager's surname:

>>> Anderson

Enter the city:

>>> London

The branch manager in London is Anderson

*The program
uses user input.*



Brainstorm

The input() function

`input()` is a function to input data from the keyboard.

```
result = input('Phrase')
```

↑
The **result** of the
algorithm's
execution:
a variable.

↑
The **algorithm**
reading the
data.

↑
A hint for users.



Brainstorm



The input() function

`input()` is a function to input data from the keyboard.

```
add_services = input('Price of additional services:')  
total = 2500 + add_services  
print('Total price:', total)
```

What will the program print?



Brainstorm

The input() function

`input()` is a function to input data from the keyboard.

```
add_services = input('Price of additional services:')  
total = 2500 + add_services  
print('Total price:', total)
```



Price of additional services:

>>> 300

[2:0] unsupported operand type(s) for Add: 'int' and 'str'

*Looks like there
is an error in
the program!*

*Can you guess
what it is?*



Brainstorm

The input() function

`input()` is a function to input data from the keyboard.

```
add_services = input('Price of additional services:')  
total = 2500 + add_services  
print('Total price:', total)
```



```
Price of additional services:  
>>> 300
```

```
[2:0] unsupported operand type(s) for Add: 'int' and 'str'
```

The **result** of the `input` function's execution is a **string**, not a number. A computer does not know how to add up integer and string values.



Brainstorm

Int() and str() functions

`int()` and `str()` functions are used to switch from one data type to another.

```
add_services = input('Price of additional services:')  
add_services = int(add_services)  
total = 2500 + add_services  
print('Total price:', total)
```

What will the program print?



Brainstorm

Int() and str() functions

`int()` and `str()` functions are used to switch from one data type to another.

```
add_services = input('Price of additional services:')  
add_services = int(add_services)  
total = 2500 + add_services  
print('Total price:', total)
```



```
Price of additional services:  
>>> 300  
Total price: 2800
```

The computer knows how to add up integer values.

The program works correctly.



Brainstorm

Int() and str() functions

`int()` and `str()` functions are used to switch from one data type to another.

```
add_services = int(input('Price of additional services:'))
total = 2500 + add_services
print('Total price:', total)
```



```
Price of additional services:
>>> 300
Total price: 2800
```

There is a shorter way to write the program.



Brainstorm

Int() and str() functions

`int()` and `str()` functions are used to switch from one data type to another.

```
add_services = input('Price of additional services:')  
add_services = int(add_services)  
total = 2500 + add_services  
print('Total price:', total)
```



```
Price of additional services:  
>>> 300  
Total price: 2800
```

The str() function switches data to the string data type.

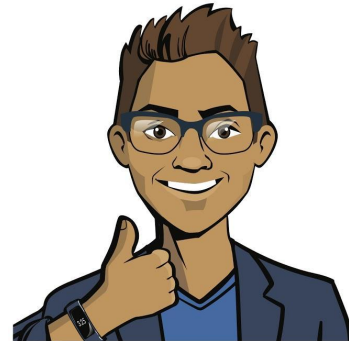
We will discuss it in detail next time.



Brainstorm

Conclusions:

1. The `input()` function is used to input data from the keyboard.
2. The result of the `input()` function's execution is a string variable.
3. To go from the string type to the integer type, you need to use the `int()` function.



Brain
storm

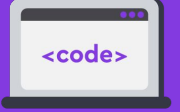


Travel agency: Optimization



Do tasks on the platform

➡ “Travel agency: optimization”



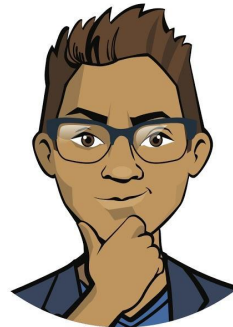
Travel agency:
optimization

End of the workday



To complete today's work, pass a technical interview

1. What is a variable? What data types can be stored in a variable?
2. How do we go from the string data type to the integer data type? Where is this used?
3. What do we use the `input()` function for?



Cole
Senior Developer



Emily,
Project Manager



Wrapping up
the workday

Congratulations on completing the workday!

Today you have learned:

1. How to make your code universal by using variables.
2. How to read input data.
3. How to change the type of the input received.



Wrapping up
the workday

Performance review

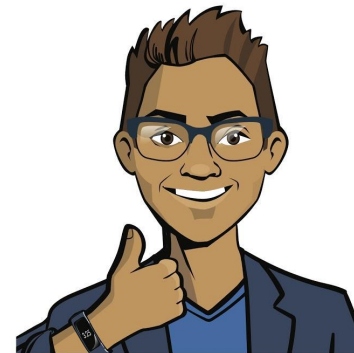
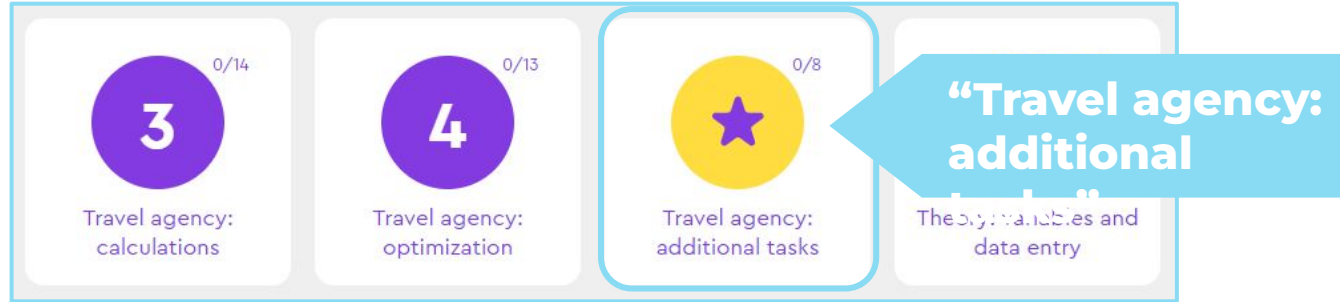
Share with your colleagues:

1. What was the best thing you managed to do?
2. What didn't work out the way you wanted?
3. What should you do next time to ensure success?



Wrapping up
the workday

Additional tasks to improve efficiency

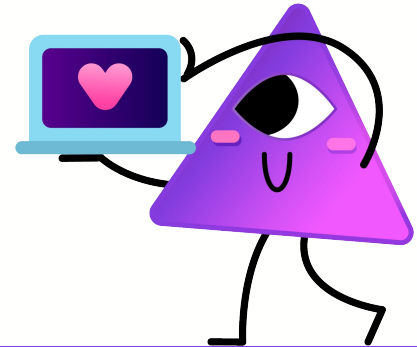


Wrapping up
the workday



Hall of Fame

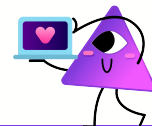
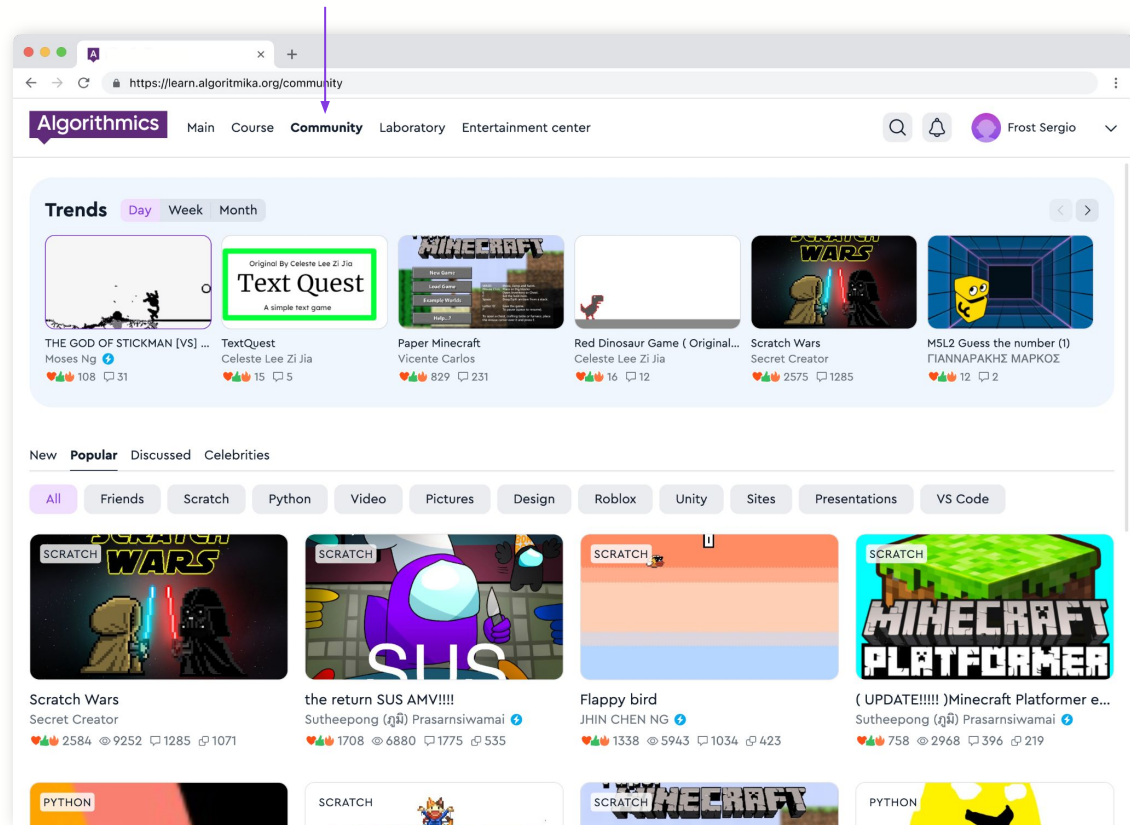
Share projects with other students



Hall of Fame

A space where our guys share projects, games, and ideas with each other.

In the Hall of Fame, the guys communicate and compete with each other

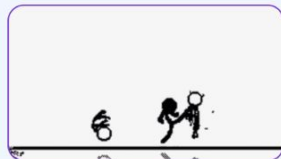


Trends

Day

Week

Month



THE GOD OF STICKMAN [VS] ...

Moses Ng

👍👍👍 108 💬 31



TextQuest

Celeste Lee Zi Jia

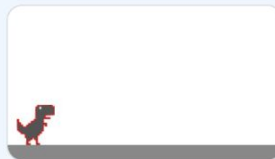
👍👍👍 15 💬 5



Paper Minecraft

Vicente Carlos

👍👍👍 829 💬 231



Red Dinosaur Game (Original...

Celeste Lee Zi Jia

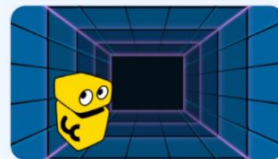
👍👍👍 16 💬 12



Scratch Wars

Secret Creator

👍👍👍 2575 💬 1285



M5L2 Guess the number (1)

ΓΙΑΝΝΑΠΑΚΗΣ ΜΑΡΚΟΣ

👍👍👍 12 💬 2

Trends

A block with new projects that are gaining popularity. The projects in this block help to understand what is happening in the Hall of Fame.

The trends reflect the projects that have the most likes and comments per day, week or month.



New **Popular** Discussed Celebrities

All Friends Scratch Python Video Pictures Design Roblox Unity Sites Presentations VS Code



Scratch Wars
Secret Creator
👍👍👍 2584 🗨️ 9252 💬 1285 🔄 1071



the return SUS AMV!!!!
Sutheepong (ភ្នំ) Prasarnsiwamai 🔒
👍👍👍 1708 🗨️ 6880 💬 1775 🔄 535



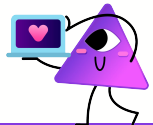
Flappy bird
JHIN CHEN NG 🔒
👍👍👍 1338 🗨️ 5943 💬 1034 🔄 423



(UPDATE!!!!)Minecraft Platformer e...
Sutheepong (ភ្នំ) Prasarnsiwamai 🔒
👍👍👍 758 🗨️ 2968 💬 396 🔄 219

Viewing projects

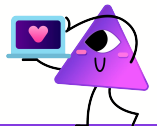
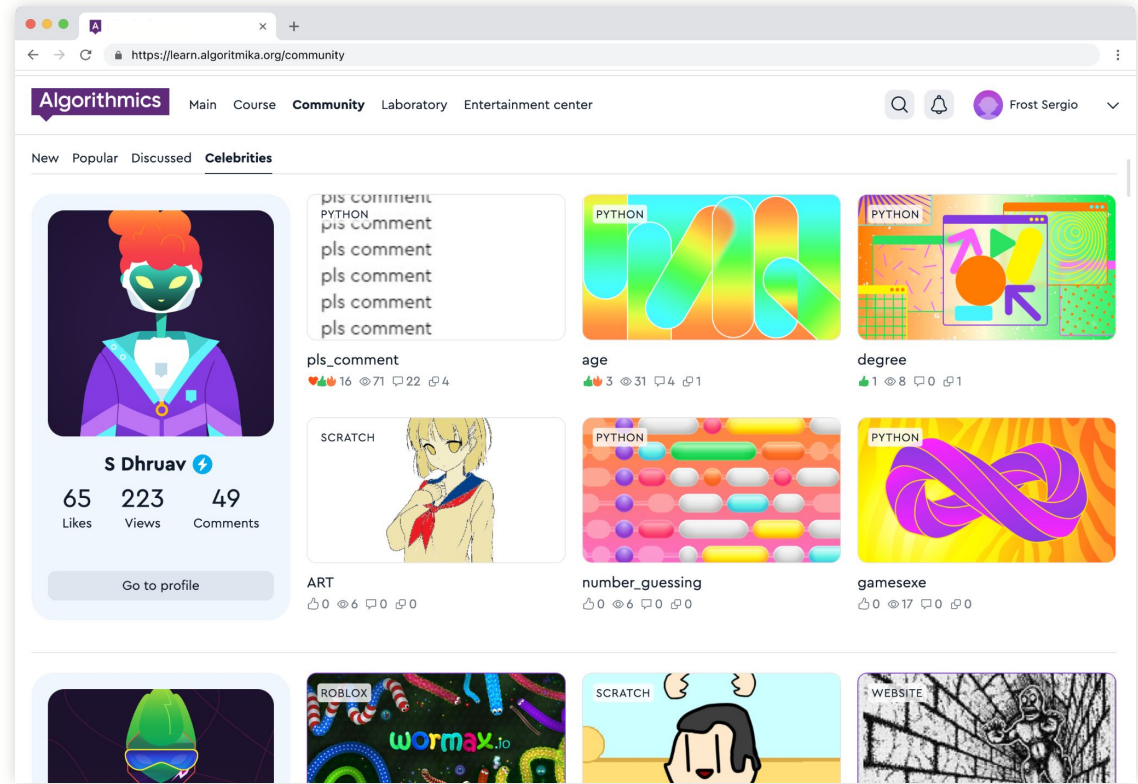
Projects are displayed by newness, popularity and number of comments, and you can also view projects of "Celebrities", friends, and projects on individual subjects



Celebrity

Students who make cool projects and win competitions receive a special status: "Celebrity".

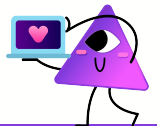
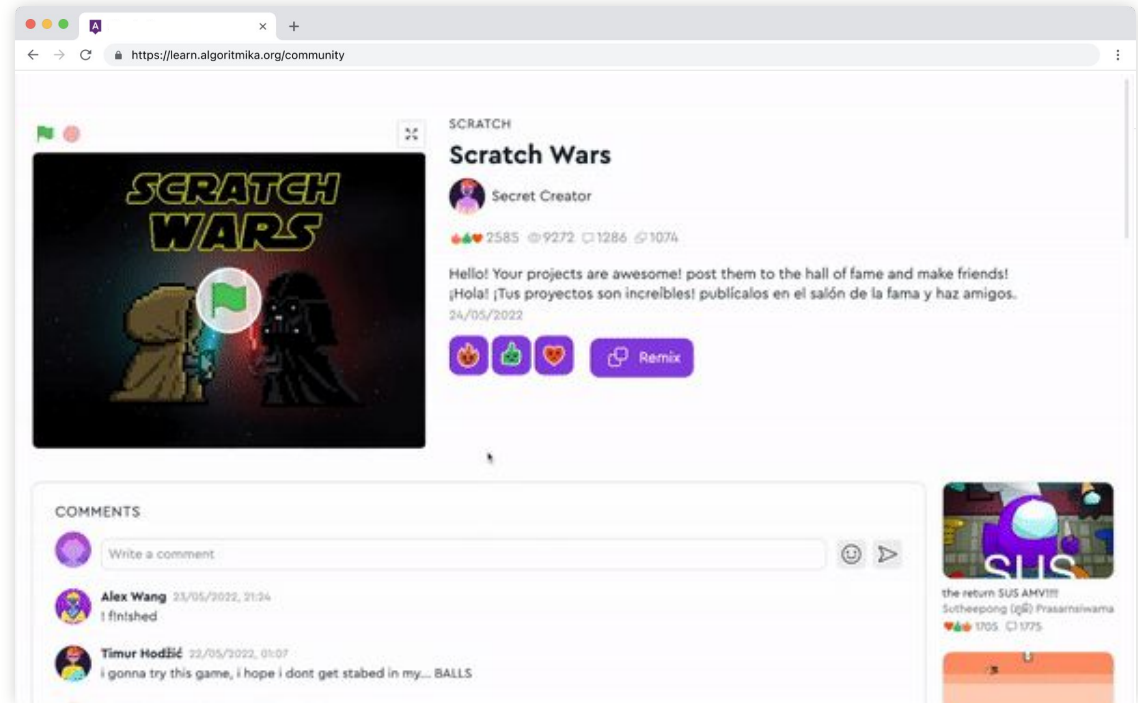
A student who has received celebrity status receives a blue badge .



Projects

See the work of other students, play games.

Put reactions, share comments and discuss projects.



Hall of Fame rules

When you publish projects or write comments, make sure that you don't break the rules:



Politeness

Do not scold or insult other students.



Honesty

Do not publish false information about events or people.



Peace and friendship

Don't add scenes of violence or crimes to your projects.



No prohibited materials

Do not publish indecent pictures, videos, text, etc.

For violating the rules, your access to the Hall of Fame will be restricted:

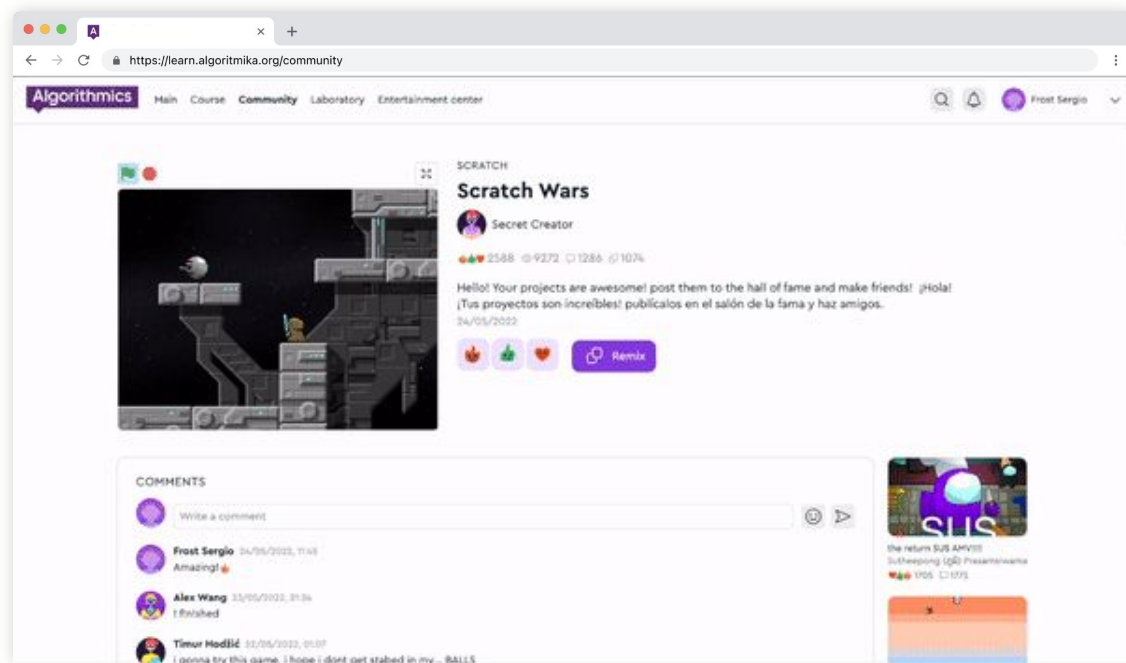
- for one week for the first violation,
- for two weeks for the second violation,
- for one month for the third and next violations, and we will also write to parents



Full-screen project

It's easier to interact with projects if they are open in full screen mode.

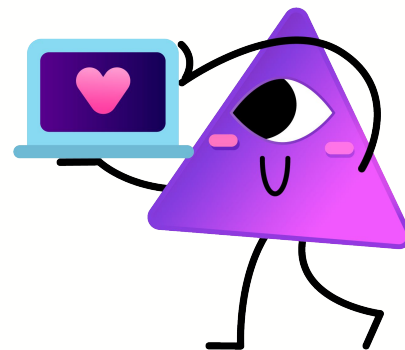
To open the project, click on the arrow icon





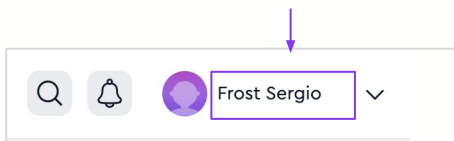
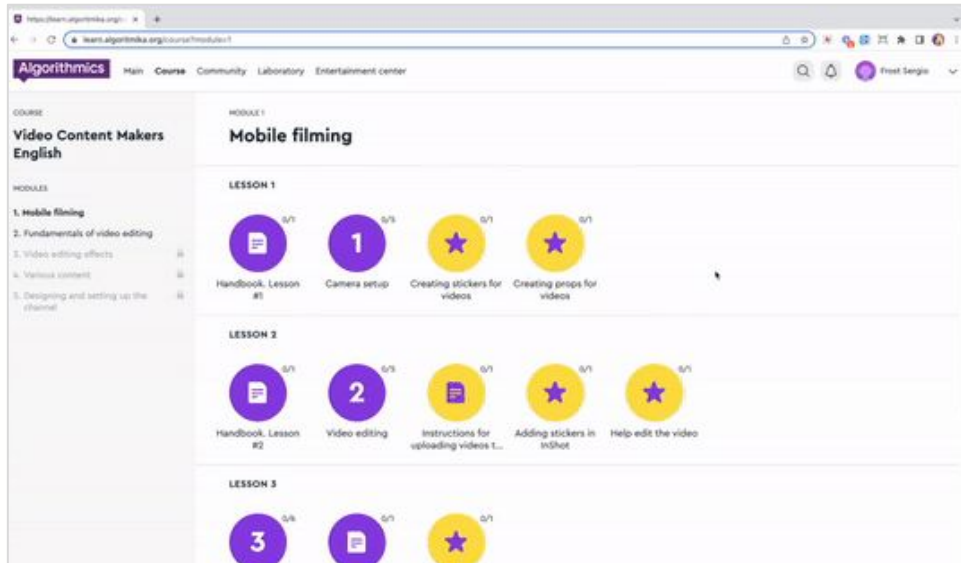
My class

Messages from the teacher and projects of classmates

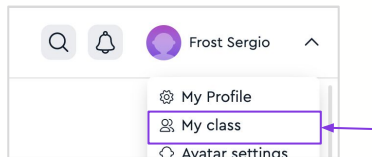


My class

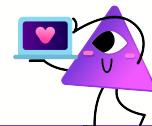
The space in your classroom where you can publish and discuss projects with classmates, and view messages from the teacher.



1. Click on your name in the upper right corner



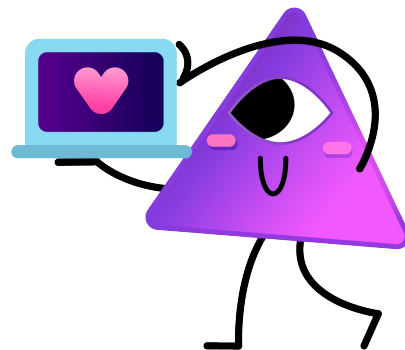
2. Click on the item "My class"





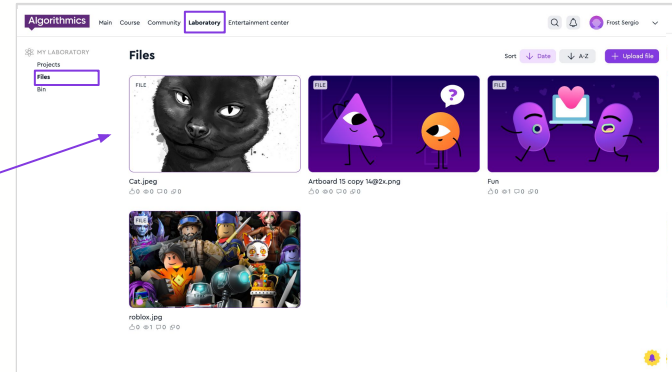
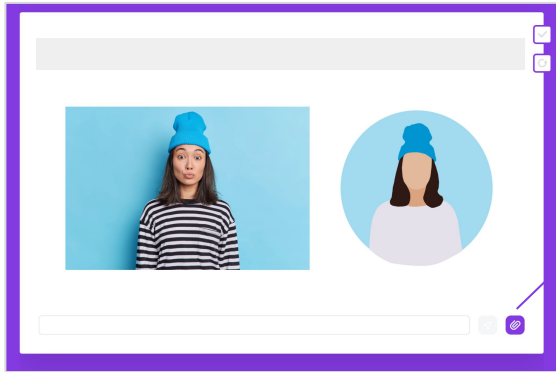
Laboratory

Storing and viewing your projects



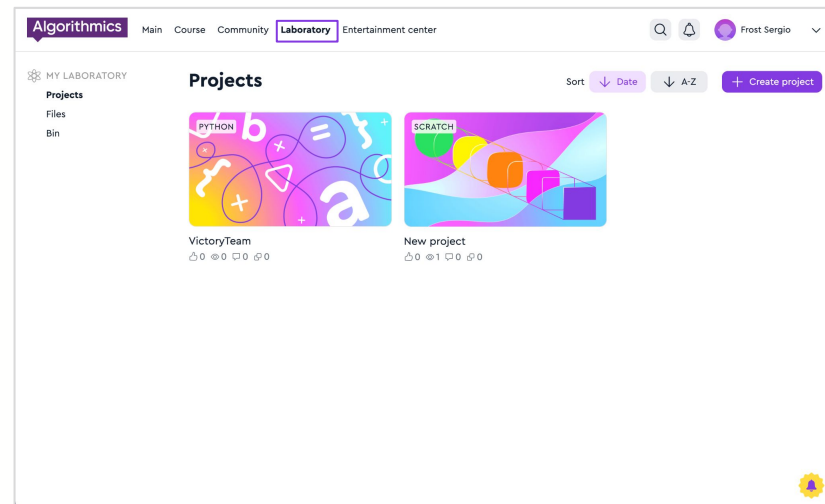
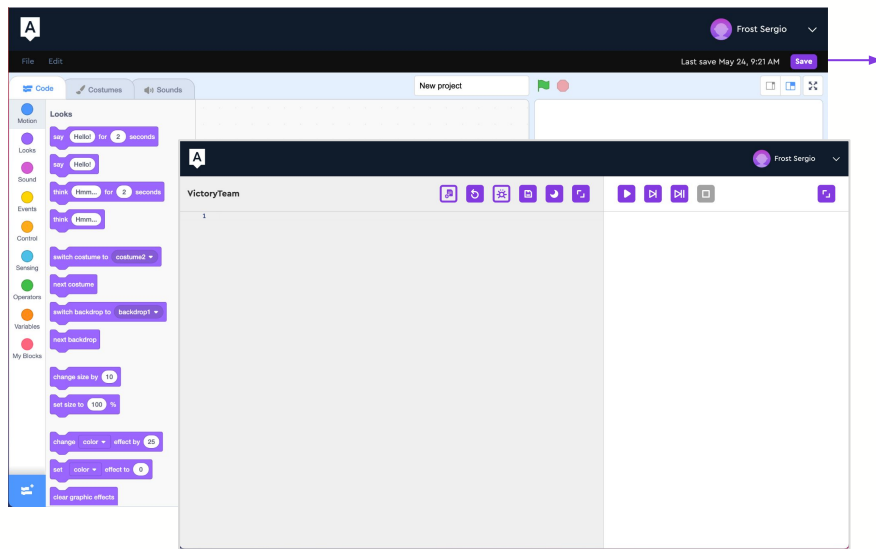
Your solutions are stored in the Lab

The files that you uploaded to the task on the platform go to the "Lab" on the "Files" tab.



Your solutions are stored in the Lab

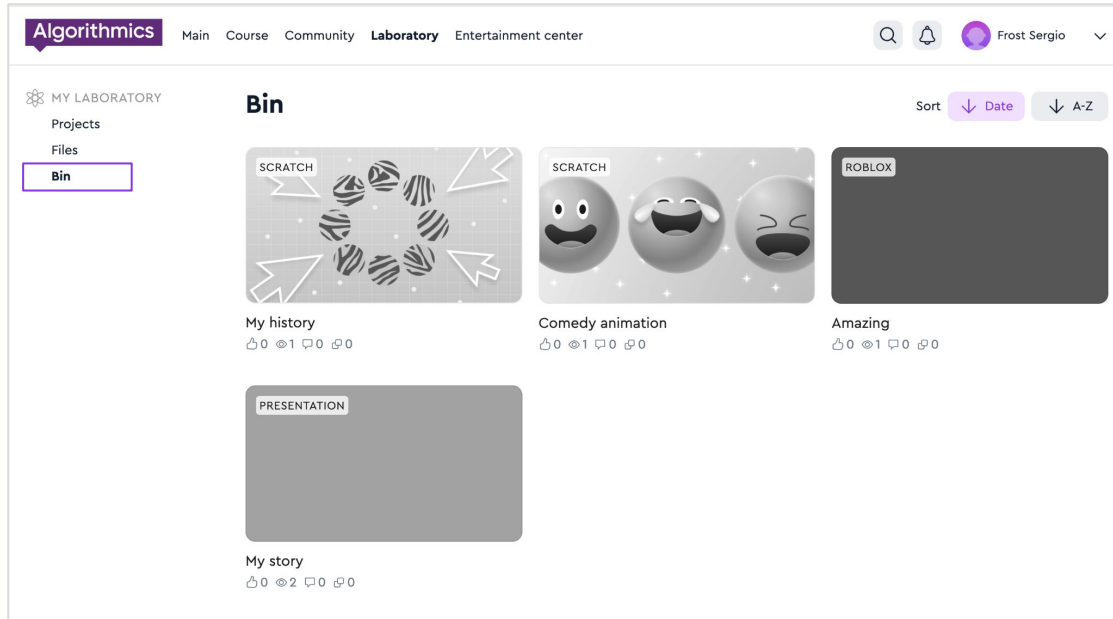
Scratch and Python solutions go to the "Lab" on the "Projects" tab immediately after saving.



Hall of Fame



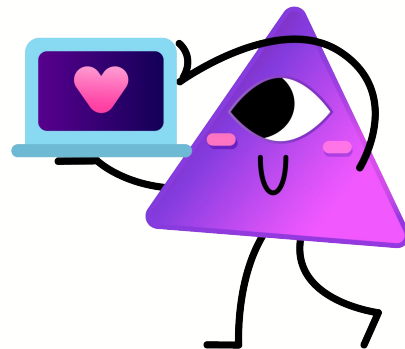
Deleted projects end up in the "Trash"





Laboratory

Publishing Python projects

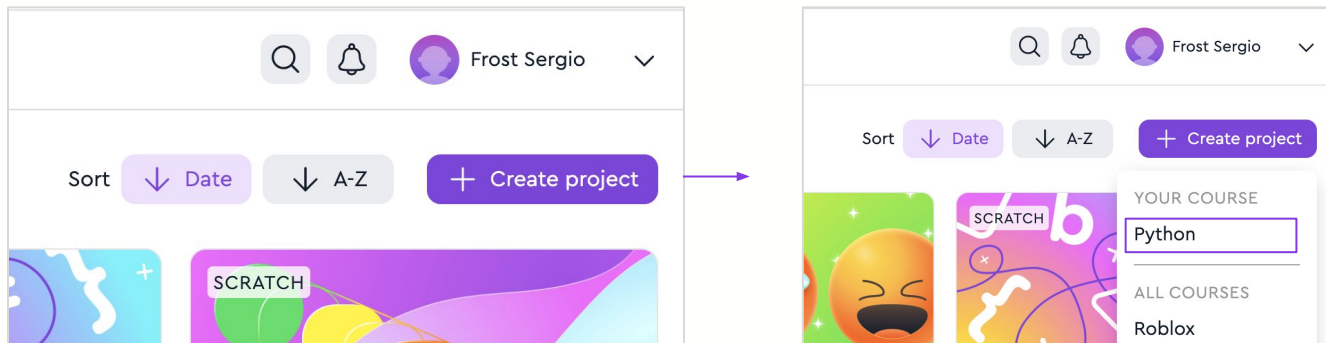


Create a project in the Lab

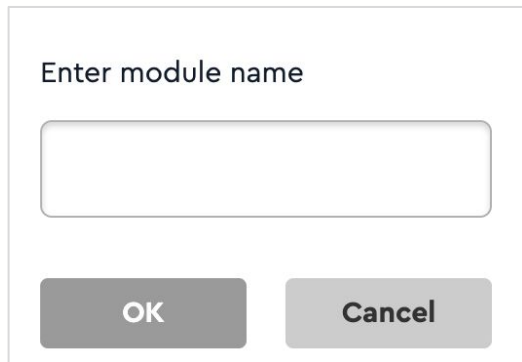
1. Go to the "Laboratory" page



2. Click on the button "+ Create project" → Python



Add a name

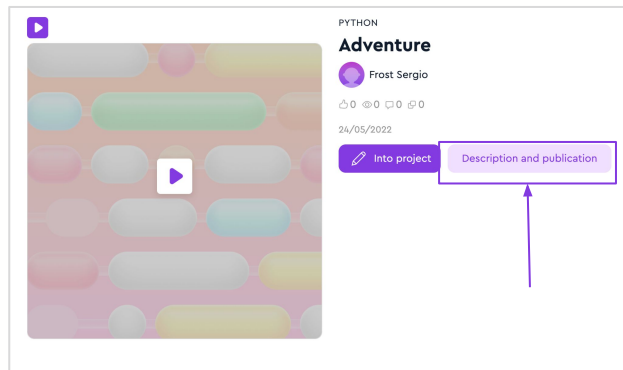


Enter module name

OK Cancel

3. Enter the project name

Use only Latin letters, numbers and the sign "_" The name should not be the same as other names of projects on Python



4. Click "Description and publication"



Enter a description and publish

Description and publication

TITLE

Saved 24/05/2022, 10:34

Adventure

DESCRIPTION

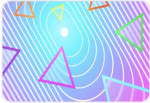
PROJECT IMAGE

<

>



Replace





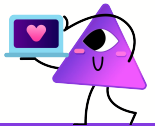
Delete project

5. Enter the name and description

6. Select and upload an image

7. Publish your project to the Hall of Fame or Class

Laboratory

A small, purple, triangular cartoon character with a single large eye and a smile. It is holding a small laptop computer in its right hand. The laptop screen displays a red heart icon. The character is standing on a thin black line representing its legs.